S1 Text. Supplementary scientific methods

Longitudinal motor-imagery EEG decoding: Frozen baselines, supervised recalibration, and task-measurement reliability

This supplement defines the completed frozen analyses and the separately labeled post-observation sensitivities reported in the PLOS ONE submission draft. It introduces no model, adaptation gate, new fit or repeated query inference. The original scientific records and all incomplete-window exclusions remain intact; operational authorizations, storage receipts and fault recovery are retained separately in the repository audit records.

S1. Source identity, chronology and eligible windows

The source was the Kumar2024 derivative at NEMAR `nm000177/v1.0.5`, CC BY 4.0, bound by the existing 2,705-file selection manifest. The analysis covered 18 published participant identities, six ordered sessions, 360 EDF recordings and 7,200 event identities. Source size/checksum, local SHA256, annotation/event agreement, channel order, sampling rate, units and frozen numerical window hashes were checked. Complete raw and processed windows had no exact numerical duplicates in the frozen split audit. These checks do not independently establish biological identity or the accuracy of derivative-reported calendar timestamps. Session order and within-session run order were checked against the source acquisition metadata.

Development IDs were sub-1, sub-4, sub-7, sub-10, sub-13 and sub-16. The fixed remaining IDs

were sub-2, sub-3, sub-5, sub-6, sub-8, sub-9, sub-11, sub-12, sub-14, sub-15, sub-17 and sub-18. Assignment preceded performance calculations and used every third numeric published ID, starting at one. Later analysis of previously inspected test identities remains exploratory.

Exactly five seconds and 2,560 samples at 512 Hz were required for a complete exported cue-anchored task-label window. Sample completeness does not certify uninterrupted imagery or isolation of the active task phase; S7 details this distinction. The 44 partial events remained in source/split records and were excluded from tensors without padding, replacement or selecting another run. The included role counts were:

Role	Fixed whole-run assignment	Included windows across 18 IDs
Source training	ses-0, run-0 and run-1	720
Source validation	ses-0, run-2	360
Source query	ses-0, run-3	360
Target support	ses-1 through ses-5, run-0	1,786
Target query	All other target-session runs	3,930

Each person had 40 training and 20 validation windows. The fixed first support run allowed at most 20 events, at most 10 per class, before partial exclusion. Among 90 target sessions, 14 had 19 included support events and 76 had 20. There was no replacement or accumulation of support from other target sessions. The query inventory retained all 4,320 event identities, including 30 partial exclusions, yielding 4,290 included queries. The saved-result comparison verified the same trial IDs, labels, sessions, runs, roles and included-window hashes for every CPU method and every EEGNet seed.

S2. Epoch-local preprocessing and descriptive coordinates

The channel order was F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, C3, Cz, C4, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8 and POz. Labels were left hand = 0 and right hand = 1. MNE numerical input was in volts; the native physical metadata used microvolts. Each epoch independently underwent:

1. Common-average reference over all 22 channels at each sample.
2. Fourth-order SciPy Butterworth SOS band-pass filtering at 8–30 Hz with `sosfiltfilt` and its default odd padding.
3. `resample_poly(up=1, down=4)` with its default Kaiser-5 window and constant padding, yielding 22 × 640 float32 values at 128 Hz in volts.

These operations fitted no state across epochs. Zero-phase filtering is an offline convention. A raw within-window peak-to-peak value above 250 µV in any channel generated a fixed quality flag. All 1,512 flagged complete windows remained among the 7,156 primary windows. No expert artifact certification or verified reconstruction of the acquisition CPz reference was claimed.

The predefined power description used Welch estimates with a Hann window, 256-sample segments, 128-sample overlap, constant detrending, density scaling, one-sided output and mean averaging. Mu power integrated 8–13 Hz and Beta power 13–30 Hz by trapezoidal integration including band-edge bins. Features were `log10(max(power, float64_tiny))`, ordered channel-major with Mu then Beta, giving 44 features. They are absolute task-window powers in V^2 before the log transform, not baseline-normalized ERD.

Covariance descriptions used a fixed 21 × 22 Helmert basis to remove the CAR null direction. Epochs were time-centered, covariance used the `n−1` denominator, and trace-normalized covariance was shrunk as `0.9*C + 0.1*I/21`. The source-training-only affine-invariant Riemann mean used an arithmetic initializer, unit step, at most 100 iterations and tolerance `1e-10`. Other roles supplied

distances to that fixed reference, never a refit. A source-training StandardScaler and full-SVD PCA retained four components without whitening. Primary preprocessing and decoder fits did not change in the fixed quality-flag, count-matched or C3/Cz/C4 descriptive sensitivities.

For the pre-frozen drift associations, the covariance measure was target-query mean distance minus source-query mean distance. The task-axis measure was the one-dimensional Wasserstein distance between target and source query projections onto the fixed source-training right-minus-left power direction. Target labels did not construct that drift variable; labels were used for BA and explicitly supervised retrospective class descriptions. These remaining-cohort associations were descriptive analyses, not development-only reliability probes or policy selection.

S3. Decoder definitions and label budgets

S3.1. CPU baselines

CSP used four components, Ledoit–Wolf regularization, log mean-square features and `norm_trace=False`; LDA used `solver="lsqr"` and automatic shrinkage. The baseline seed was 42 and there was one prespecified configuration. Source-only CSP/LDA fitted source training and then froze both stages. Frozen-CSP/LDA recalibration reused source filters and refitted LDA on source training plus current support. Full refitting fitted both CSP and LDA on that same union. Every fit recorded exact trial IDs; validation and query were forbidden from fitting or selecting the CPU pipeline. Source-query predictions used the same source model for all three method labels.

S3.2. Source-only EEGNet

The official `braindecode.models.EEGNet` implementation used F1=4, D=2, F2=8, temporal kernel 64, depthwise kernel 16, mean-pool sizes 4 and 8, ELU, dropout 0.5, automatic final-convolution length and spatial max norm 1.0. Batch normalization used momentum 0.01, affine parameters and epsilon 0.001. `final_layer_with_constraint=false` and `norm_rate=0.25` were retained exactly as recorded in the executed configuration.

For each participant, channel mean and standard deviation were fitted over source-training trials and time only, with a `1e-8` V scale floor. The optimizer was Adam: learning rate 0.001, betas (0.9, 0.999), epsilon `1e-8`, zero weight decay, no AMSGrad or scheduler. Training used unweighted cross entropy, no label smoothing, batch size 16, shuffled training rows, `drop_last=false`, `num_workers=0` and at most 100 epochs. The source-validation mean cross entropy selected the checkpoint, with patience 15, minimum improvement zero and the earliest epoch on ties. The complete best state was restored. All seeds 20261012, 20261013 and 20261014 were retained; no favorable seed was selected.

The 54 completed source fits used 2,871 epochs and 8,613 optimizer steps. Eleven reached the 100-epoch cap; reaching that cap was not relabeled as a patience termination. Validation BA/CE was descriptive because validation selected epochs. It received no new population interval or hypothesis test.

For the fixed checkpoint query evaluation, original package/checkpoint bytes and embedded source context were verified before loading with `torch.load(map_location="cpu", weights_only=True)`. Evaluation mode, disabled gradients and inference mode prohibited optimizer, parameter, batch-normalization and normalization updates. The original state digest was recorded before and after each model's predictions. Labels supported only the fixed metrics; class prediction was untuned logit argmax. The original source CE replay absolute tolerance remained

`1e-7`. Independent saved-result audits checked the recorded states and arithmetic without a second model forward.

S3.3. Comparison boundary

Pipeline	Source estimator/optimizer labels	Additional source-validation labels for selection	Current support labels for fitting
Source-only CSP/LDA	40	0	0
Frozen CSP + LDA recalibration	40	0	19/20
CSP/LDA refit	40	0	19/20
Always-frozen EEGNet, each seed	40	20	0

The CPU validation run was not used for model selection or fitting. EEGNet therefore had different source-label use, and only the CPU updates used target support. Equal query identities do not establish equal information or compute budgets. The two CPU update methods shared exact labeled fitting IDs, although updating both stages and updating LDA alone involved different computations. The original full-refit versus frozen contrast deliberately added current support; it was not an equal-target-label architecture comparison.

S4. Endpoints, resampling and evidence hierarchy

BA was the mean of left and right recall. Confusion counts, recalls, predicted class proportions and single-class prediction indicators accompanied each session. CPU target endpoints first averaged five sessions per participant. EEGNet first averaged all three seeds within participant/session, then averaged five target sessions per participant. Source query remained separate. Cohort means weighted participants equally, never pooled trials or treated seeds and sessions as additional independent people. The six development participants and 12 primary participants were analyzed separately; combining all 18 was a secondary exploratory analysis.

The primary contrast remained full CSP/LDA refit minus source-only CSP/LDA in the remaining 12, with paired participant resampling. Mean/BA descriptive intervals used 10,000 draws with seed 42 and the 2.5th/97.5th percentiles. The saved-query comparison and decoder endpoint summaries used lexicographic participant-ID order. The six-person reliability bootstrap used numeric order 1,4,7,10,13,16, averaged source folds or support sessions within person first, and used `default_rng(42)` for 10,000 six-ID samples with replacement. Missing participant values made the entire reliability interval unavailable rather than silently changing its participant denominator.

The original descriptive Spearman drift intervals instead used **2,000 participant-block bootstrap draws**, seed 42. Each selected participant contributed all its session rows together. They did not use 10,000 independent session or trial samples. Both original drift measures retained 2,000 valid block draws. The difference between these two resampling designs is preserved.

No EEGNet-minus-CPU contrast interval or hypothesis test, or formal test against 50%, was added. The post-review exploratory CPU sign-flip checks and their four-test Holm adjustment are reported separately in S11. The saved comparison reproduced the original CPU primary contrast, +7.42 percentage points [2.49, 12.48], with nine positive and three negative participant differences. The original full-refit minus LDA-only secondary contrast was +5.47 percentage points [2.17, 8.91]. Its

target-label budget was the same for the two update methods; it was not promoted to a primary endpoint. LDA-only minus frozen remained uncertain at +1.95 points [−0.98, +5.09]. All subsequent same-dataset interpretation remained exploratory after prior test exposure. No task-aware policy benefited from looking at these query outcomes.

S5. Development-only measurement reliability

S5.1. Fixed low-dimensional power probes

The candidates were standardized 44-dimensional Mu/Beta power, its source PCA4 coordinates and six C3/Cz/C4 Mu/Beta coordinates. Only the six development IDs' source training and preassigned current support entered the calculations. Source-run probes trained on run-0's 20 rows and evaluated run-1's 20, then reversed the roles. Each run contained ten events per class. StandardScaler and PCA for a fold were fitted to its TRAIN20 alone. The diagnostic classifier used the right-minus-left class-centroid axis and the midpoint threshold. It did not reuse a transform fitted on the held-out source run.

The exact Cartesian set of 20 × 20 circular label phases was shared across the two source folds and three candidates, including identity. Label-dependent centroid directions and midpoints were recomputed for controls; label-independent fold scales/PCA could be reused. The 14,400 phase-control fold/candidate evaluations included the 36 identity/observed probes; these were auxiliary measurements, not replacements for the primary decoder baseline. All candidates remained; the highest point estimate did not select a winner.

S5.2. Supervised SOURCE20 CSP transfer probes

The completed signed-score run02 fitted CSP and LDA anew on each TRAIN20 fold. Because CSP is label-dependent, every training-label phase refitted both stages. For each participant/fold, 20 training-label phases produced 20 cached models and their held-out predictions. Crossing those predictions with the 20 held-out label phases yielded all 400 joint `(run0_shift, run1_shift)` pairs. Held-out labels never entered fitting, so caching across held-out phases preserved the exact full-pipeline controls.

Across six IDs, this produced 240 fitted pipelines—240 CSP plus 240 LDA fits, including the 12 identity-fold pipelines—4,800 cached held-out trial predictions, 4,800 fold/phase BA evaluations and 2,400 participant/phase-pair twofold means. All twelve source runs had 20 unique circular label assignments. The observed value was the identity phase pair `(0,0)`. All 240 attempted pipelines completed; the implemented expected singular-fit failures would have remained explicit unavailable values, not exclusions of participants or controls.

S5.3. Frozen SOURCE40 coordinates and signed scores

Current support used the six existing SOURCE40 baseline CSP/LDA models without a new decoder fit. For an exported cue-anchored window `X`, the four CSP features were the **natural logarithms** of mean squared projected amplitude:

`f_j(X) = ln(mean_t((W_j X_t)^2)), j=1,...,4.`

The executed consumer rejected nonpositive/nonfinite power rather than inventing features. The vector diagnostic fitted a four-coordinate StandardScaler on SOURCE40 features only, using population variance (`ddof=0`), and applied that fixed transform to support. Its fixed source direction was `d_source = mean(z|right) − mean(z|left)` and its unit direction was `u_source = d_source / ||d_source||` when estimable.

Within each support class, trials were chronologically ordered by acquisition time, start sample and source row. The first `floor(n_class/2)` entered the first half and the remainder entered the second; corresponding class halves were combined. For a half direction `d_h`, the diagnostics retained source/full support cosine, half-to-half cosine and the signed axis contrast

`A_h = (d_h · u_source) / SD_SOURCE40(z · u_source).`

This contrast has **no factor of one half**. Its source projection SD used `ddof=1`. Zero directions, missing classes and non-estimable scales remained explicitly unavailable.

The separate LDA diagnostic used original unstandardized CSP log features, saved coefficient/intercept and classes `[0,1]`:

`s = f @ w + b; prediction = right if s > 0, otherwise left.`

The intercept and zero threshold remained unchanged; support was neither centered nor used to fit a score scale. With `SD_s = SD_SOURCE40(s)` using `ddof=1`, each half retained the balanced true-class signed margin

`M_h = (mean(s|right,h) − mean(s|left,h)) / (2*SD_s).`

The factor of one half is part of this score-margin definition. Separate right margin `mean(s|right,h)/SD_s`, left margin `−mean(s|left,h)/SD_s`, score location, recalls, predicted proportions, BA and collapse were also saved. The vector-axis contrast and LDA margin are distinct measurements. Neither is a calibrated probability or a validated neural-information measure. A positive half-product can arise from two negative margins, and a positive class contrast can coexist with both classes on one side of the frozen threshold.

S5.4. Support controls and denominators

Each current support run supplied all 19 or 20 circular label phases, including identity, with source features, axes, scales and LDA fixed. The primary control recomputed classwise chronological halves after every phase. A separate sensitivity retained the original half indices while rotating labels. For an estimable observation, the descriptive upper-tail fraction was `count(available control >= observed) / count(available control)`, retaining ties and identity without a plus-one adjustment. Attempted, available, unavailable and unique-assignment denominators were recorded explicitly. Unverified stationarity, cyclic boundaries and cue-order dependence prevent treating these fractions as formal permutation p values.

For CSP run02 there were 596 support phases per half method; all 596 were available for each of four saved control metrics—source/full-support cosine, half-to-half cosine, vector half-contrast product and LDA half-margin product— with zero unavailable. For the three power candidates there were 1,788 phases per method, all available for each of three metrics; the two methods contributed 3,576 method/phase evaluations. Phase counts do not add participants. No sign flip, candidate, confidence cutoff or adaptation threshold was selected from the results.

S6. Retained outcomes and numerical-validation scope

The neural query analysis retained 16 collapsed seed/session outcomes across seven source-model jobs and seven participants: 14 target and two source outcomes. The frozen CPU model retained three target-session collapses; its two updates retained zero. These denominators differ and their raw counts are not a matched statistical comparison. All participant decreases and all three neural seeds stayed in the aggregates. The five collapsed source-validation fits and eleven 100-epoch terminations also remained in the source records.

CSP run02's observed source twofold mean BA was 62.9167% [52.5, 71.6667]; the participant

mean of five support half-cosines was 0.295325 [0.032041, 0.573003]. Among 30 support sessions, signed score half-margins were both positive in 19, both negative in two and opposite in nine; five sessions had frozen-score class collapse. These are calibration-pool measurements, not independent target-query performance of an adaptation rule.

The completed CSP measurement retained the original scientific definition. Its saved-feature numerical check reconstructed 629,137 arithmetic fields with maximum difference zero; it did not independently recompute EEG projections or refit estimators. Earlier incomplete or differently specified diagnostics are not pooled with these measurements. Their original files and failures remain in the separate audit history rather than being rewritten as successful attempts.

S7. Exported cue windows and unresolved within-trial task phases

The released intervals are **cue-anchored exported task-label windows**, rather than five seconds certified to contain uninterrupted motor imagery. Three questions are distinct: whether all samples of an interval are available, whether its start corresponds to the intended marker, and whether its complete span lies within active imagery or feedback. The first is established for the 7,156 included intervals. The documented cue convention supports the second at the derivative/export level. The third remains unresolved at trial level.

The original authors' codebook distinguishes class cue markers 769/770 from subsequent task-start markers 7691/7701 [1]. The preserved official MOABB Kumar adapter maps the former pair to left/right hand and declares an interval of [0,5] seconds. NEMAR identifies MOABB conversion and preserves corresponding onset/sample/class rows. Its metadata reports MOABB `1.5.0dev0`, but the exact historical converter commit/options and original GDF marker arrays have not been

independently replayed. A current adapter is evidence of the documented convention, not proof of every historical conversion operation.

The original Methods describe one-second fixation, a 1.5-second class cue, then a five-second offline task period or online feedback lasting at most seven seconds; a two-second result display and 1.5-second rest follow [2]. Online evidence-threshold crossing or timeout can terminate the active period. Consequently, a nominal five-second interval beginning at the class cue contains cue presentation and at most the following 3.5 seconds of the task phase, assuming the described phase schedule. Earlier online termination may introduce post-termination signal, but its presence and duration cannot be assigned to individual intervals from the retained labels. We do not claim that every online interval includes result display or rest.

The source codebook describes 32766 as recording boundary triggers and 1000 as trial/rest start. For 7692/7702 it describes offline trial end and online timeout/incorrect-hit events. The online 7693/7703 description is literally “Cue Left/Right Delivered”; we do not silently reinterpret it as a correct hit. The authors’ versioned task-bounded extraction function starts at 7691/7701 and terminates at the subsequent 7692/7702 or 7693/7703. This supports the distinction between their task-period markers and the class-cue export, without equating either annotation convention with evidence of sustained imagery physiology.

The 360 saved event tables contain `onset`, `duration`, `trial_type`, `value` and `sample`, but not the complete within-trial marker sequence. Each of the 44 partial intervals is the twentieth and final retained class event of an online recording and ends at its previously audited EDF sample boundary. Their identities, lengths and exclusion reasons remain unchanged. This demonstrates recording-boundary truncation of the stored interval, **not 44 verified early task terminations**, and does not by itself establish an indexing error. No retained interval overlaps the next retained class-event onset; absence of this overlap still cannot demonstrate continuous imagery within a window.

S7.1. Sources actually searched and the availability boundary

We checked all 360 retained NEMAR event tables (7,200 rows), their official checksums, and the saved checks comparing the complete derivative EDF annotation stream with those tables. The TSV fields are `onset`, `duration`, `trial_type`, `value` and `sample`; the only class labels are left/right hand and the only values are 1/2. They contain no separate task-start/end or outcome records. The EDF statement relies on the prior hash-matched audit and script, not a fresh read of the signals in this revision.

We also inspected the original-author codebook and extraction function, the complete public source-repository tree, NEMAR's complete source-file manifest, conversion metadata and saved backup inventories. The full manifest lists 72 offline and 288 online original bar-task GDF files, in addition to racing data. NEMAR describes these source files as unmodified upstream originals. None of the GDF files was part of the frozen derivative selection; their embedded event arrays and their mapping to the analyzed trial/sample identities were not parsed in this review. The public author repository also contains example GDFs, which were not opened here. No standalone target-run POS/TYP export was identified within the inspected repository tree and text files.

Thus the verified absence concerns the retained derivative annotation stream. It is not a claim that the original public distribution lacks the markers or that phase reconstruction is impossible. Code illustrating task-bounded extraction does not substitute for validating the corresponding arrays and sample mapping for all analyzed trials. The search scope and unopened sources are recorded in the public revision evidence.

The minimum evidence for a future phase-specific amendment is the original bar-task `POS/TYP` marker arrays for all 360 runs, their sample rate and sample-origin convention, and a pinned converter or maintainer-confirmed mapping to the current trial IDs. Cue, task start, outcome/task end and recording boundaries must be matched without consulting decoding outcomes. Before any new

evaluation, an amendment would need to specify one anchor and duration, handling of early ends and altered support/query budgets. Existing cue-window predictions and models would remain the original analysis; they cannot be relabeled as task-only models. No re-epoching, new fit or inference was performed here. The correction to the original paper changes reported CSP numbers, not these marker definitions [3].

The original Methods were first checked through indexed publisher/PMC passages. During this revision the official Europe PMC full-text XML was successfully obtained and the relevant Methods, framework descriptions and results were read; earlier access failures remain preserved in the audit record. The complete author codebook, task-bounded extraction source and previously saved correction XML were read. The original Methods identify UT Austin protocol 2020-03-0073 and written informed consent. These are the original study's statements only. The author's institutional determination for the present secondary analysis remains unresolved; no approval, exemption or no-review conclusion is inferred from public availability.

S8. Exploratory decoding sensitivity to the fixed amplitude-QC flag

S8.1. Status, input rules and exact identity joins

The sensitivity was requested and its protocol recorded **after the original results had been observed**, but before computing these new summaries. It is not preregistration, an amendment to the primary endpoint, artifact correction or an untouched test. The original main results and 44 incomplete-window exclusions, including 30 incomplete query identities, were preserved.

Only saved predictions, the existing Boolean `qc_peak_to_peak_over_250uv` flag and frozen text metadata were used. The flag was checked against the saved maximum raw channel peak-to-peak

amplitude and the fixed strict threshold `>0.00025 V` in any channel. The threshold was not changed or selected by performance. Partial-window QC entries can be null and are never silently interpreted as unflagged complete examples. The original 1,512 flagged complete windows comprised 76/720 source TRAIN, 36/360 source validation, 36/360 source query, 478/1,786 target support and 886/3,930 target query windows. Thus 922/4,290 unique included queries were flagged and 3,368 were unflagged. Flagging is not expert artifact certification; unflagged does not mean artifact-free.

Each saved prediction was joined by exact `trial_id` with unique, consistent participant/session/run/role, original label, inclusion and QC values. Every complete query required all three CPU methods and all three fixed EEGNet seeds. Unexpected, missing, repeated or inconsistent identities caused failure rather than partial scoring. The join comprised 25,740 stored predictions for 4,290 complete queries in 108 participant/method/seed groups. A separate 4,320-row query ledger retained the 30 incomplete identities and their original exclusion reasons. All 144 reconstructed endpoint values (two values, source-query and five-session target BA, for 72 participant-by-method rows) matched the saved values exactly.

S8.2. Class eligibility and aggregation

Both `all_included` and `unflagged` subsets retain rows for each participant, method, seed and each of six sessions, plus separately labeled target-pooled and all-query-pooled scopes. Counts include true left/right, predicted left/right, all confusion cells and single-class prediction flags. Empty and single-true-class groups remain present with undefined BA and an explicit reason; unavailable values are not set to zero or chance.

Mathematical BA requires at least one example of each true class. A group with fewer than five examples in either class is flagged as low-count descriptive data but is not deleted by a new minimum-count rule. A group containing both true classes can have model class collapse; a zero-query group

has no defined collapse assessment. EEGNet BA is computed separately for every seed and then averaged over **all three** within participant/session. Counts represent unique query identities, not trials multiplied by seeds, and no voting ensemble is constructed.

The target endpoint requires all five target-session BAs and gives each equal weight. If any target session is undefined, that participant's equal-session endpoint is undefined; fewer sessions are never silently averaged. Cohort means use the explicitly listed eligible participants. Source, target-pooled and all-query-pooled endpoints have their own eligibility counts and cannot replace the primary equal-session endpoint. Development 6, remaining 12 and secondary all 18 remain separate.

Two remaining-cohort identities lose their unflagged five-session endpoint for all four methods. Sub-8 has no unflagged queries in ses-3 or ses-4 and only 28 across target sessions (16 left/12 right); its source query retains one left and zero right events, so source BA is also undefined. Sub-12 retains four target events, all left-hand, distributed 1/1/2/0/0 across the five sessions. Every target session is missing the other class. Other low-count sessions remain, including sub-8/ses-2 and sub-11/ses-2 (each 2 left/1 right), and sub-5/ses-5 (3 left/12 right). Unflagged equal-session summaries therefore include 6/6 development and 10/12 remaining participants. The eligible remaining IDs are sub-2, 3, 5, 6, 9, 11, 14, 15, 17 and 18; sub-8 and 12 remain in all eligibility and missing-class tables.

S8.3. Paired sensitivities, same-participant comparisons and influence

The original remaining 12 full-refit minus frozen contrast stays the primary comparison in the original analysis. Full-refit minus LDA-only stays secondary. Their QC-restricted counterparts are **new exploratory sensitivities**. Both use paired participant differences and 10,000 percentile bootstrap draws with `default_rng(42)`, lexicographic participant-ID ordering and 2.5th/97.5th percentiles. Fewer than two eligible participants yield no interval. This original QC analysis used descriptive intervals; the separately introduced post-review paired checks are identified in S11 and do not replace

these intervals or establish an adaptation gate.

In the unflagged remaining 10, full-refit minus frozen is +5.96 percentage points [0.99, 11.24]; full-refit minus LDA-only is +3.82 [1.11, 6.72]. The corresponding original 12-person values remain +7.42 [2.49, 12.48] and +5.47 [2.17, 8.91]. Contrasting these two cohort means directly would mix QC and changed membership. For the **same 10** eligible identities, original full-minus-frozen is +6.50 and unflagged is +5.96; the paired change is −0.54 [−1.30, +0.16] points. For full-minus-LDA-only, original is +4.53 and unflagged +3.82; the paired change is −0.71 [−2.11, +0.42] points. For these same ten participants and the unchanged fitted models, the positive unflagged contrast does not support attributing their whole average difference solely to flagged query amplitudes. It does not establish general artifact robustness or recover the missing full-cohort endpoints.

Deterministic leave-one-participant-out means are retained for both CPU contrasts without excluding anyone by observed gain. The original full-minus- frozen mean ranges from +6.22 to +8.62 points over the 12 omissions; the unflagged 10-person range is +4.61 to +7.24. The three original negative individual differences are −5.82, −0.67 and −0.20 points (sub-6, 14, 15). Small negative differences are not automatically interpreted as substantial harm or tested individual effects. These are influence descriptions, not additional independent participants or an exclusion rule.

Conditioning query evaluation on amplitude changes its class/sample composition and may select observations. The 76 flagged source TRAIN examples, 478 flagged target support examples and 36 flagged validation examples were still used as originally prescribed; no historical fit or checkpoint selection was cleaned or repeated. The analysis therefore cannot remove training artifacts, resolve task-phase uncertainty, demonstrate a reliable update rule or provide an independent replication. All previously exposed-cohort follow-up remains exploratory.

S9. Saved learning histories and the weak source-only neural reference

All 54 original metadata histories supply 2,871 consecutive recorded epochs and 8,613 optimizer steps. We extracted text histories, selection metadata and existing query metric tables; binary model members were only streamed for integrity hashing, never deserialized. No query array or EEG was read and no new forward pass, optimizer or fit was invoked. Complete curves are supplied as 2,871 text rows with participant, cohort, seed, job, epoch, training CE and validation CE; the accompanying 54-fit table retains every selected epoch and stopping reason. Empty console logs in 37 archives were not treated as proof of zero execution, and no missing curve was reconstructed.

Twelve models selected epoch 1 and 11 reached 100 epochs. Across the 36 fits for the 12 primary participants and three seeds, the corresponding counts are 8 and 10; median executed and selected epochs are 29.5 and 14.5. Final training CE is below its first value in 53/54 models (35/36 remaining), whereas final validation CE is above its first value in 29/54 (19/36 remaining). Remaining-cohort medians are training CE 0.704246 initially and 0.596543 finally; validation CE 0.693048 initially, 0.691847 at the selected epoch and 0.692351 finally. These are repeated-model descriptions, not participant-level intervals. Training CE is accumulated in training mode while parameters change within each epoch; validation CE uses the post-epoch model in evaluation mode. Dropout and batch-normalization mode also differ. The loss difference is consequently not a calibrated generalization-gap measurement or proof of one failure mechanism.

In the 12-participant primary cohort, mean BA on the held-out source query was 52.9167%, compared with 51.5783% over later target sessions, a descriptive difference of −1.3384 percentage points. Weak discrimination is already present in the source session. The target score alone therefore does not identify how much cross-session drift contributes to performance. Source query has only

20 trials per participant; the target endpoint averages five sessions. These different sampling precisions preclude treating the mean difference as an isolated causal drift estimate. Selected-validation performance is not an independent query endpoint.

Every fixed seed stays in the results. Remaining-cohort target means for 20261012/20261013/20261014 are 52.42/49.94/52.37%; the average within-participant target mean BA range is 8.06 percentage points, maximum 24.62. These values do not select an advantageous seed or imply population uncertainty from 54 independent people. The finding concerns this small-source implementation, not the general suitability of deep learning for motor-imagery EEG.

S10. Scientific software versions and text-only reproduction

The CPU baseline used Python 3.11.16, NumPy 2.2.6, SciPy 1.15.3, scikit-learn 1.7.2 and MNE 1.10.2, with recorded OMP/OpenBLAS/MKL thread limits of one. The GPU source fits used PyTorch/Torchaudio 2.8.0+cu128 and Braindecode 1.2.0. Fixed checkpoint inference used PyTorch/Torchaudio 2.8.0+cpu and Braindecode 1.2.0 with the same Python/NumPy/SciPy/scikit-learn/MNE versions, one PyTorch thread, OMP/OpenBLAS limits of one and `CUBLAS_WORKSPACE_CONFIG=:4096:8`. GPU fitting and CPU inference are distinct stages; these software statements do not establish equal compute budgets.

The public QC inputs comprise the frozen 7,200-event metadata (including 44 historical incomplete identities), 25,740 saved predictions, original analysis configuration, 72 participant-by-method endpoint rows and their provenance manifest. The dictionary records left=0/right=1, `cpu_fixed` for CPU seeds, the three EEGNet seeds, roles, inclusion/exclusion reasons, original amplitude flags and input lengths/hashes. Full per-session class counts, missing groups, confusion cells, paired differences and influence rows accompany the results. No EEG samples, checkpoint parameters, private credentials or new dataset collection is included in these compact inputs. Source-derived data retain

their NEMAR CC BY 4.0 attribution; that attribution is distinct from software dependency licenses.

From a repository checkout, the following command reproduces the public saved-prediction arithmetic only, using Python 3.11 and the recorded NumPy 2.2.6. Choose a new output directory; the existing frozen results are not overwritten:

```
mkdir qc_text_reproduction_new
cp research_logs/kumar_submission_qc_20261015/run01/POST_OBSERVATION_PROTOCOL.md
 ↪  qc_text_reproduction_new/
python -B scripts/analyze_kumar_saved_query_qc_20261015.py \
  --portable-inputs research_logs/kumar_submission_qc_20261015/run01/public_inputs \
  --output qc_text_reproduction_new
```

The portable input path needs no private original checkpoint or prediction directory. The original-input computation and portable text-only replay produced the same 13 scientific CSVs byte-for-byte. These are repeat arithmetic checks, not repeat model inference. Public saved history tables can be inspected without weights. To independently validate their original archive/member hashes requires authorized possession of the 54 unmodified originals; use the portable relative-filename index and a new destination:

```
python -B scripts/summarize_kumar_saved_training_curves.py \
  --archive-index
   ↪  research_logs/kumar_submission_training_review_20261015/run01/archive_input_index.json
   ↪  \
  --archives /absolute/path/to/54_original_archives \
  --output /absolute/path/to/new_text_only_review
```

This second command reads only allowlisted metadata/text and streams binary members to SHA 256; it does not load tensor values or construct a model. Both postprocessors require explicit output arguments. Invoking either without arguments exits with usage instead of initiating fitting or evaluation. Neither command invokes the scientific training/query runners; no such rerun is needed

to reproduce the new saved-prediction summaries. The fixed protocols, code, CSV dictionaries and complete hashes provide the detailed implementation record without reproducing operational authorization or storage logs in this supplement.

S11. Post-review exploratory paired sign-flip checks

These checks were introduced in response to review after the original results were known. A computation plan and a four-comparison Holm addendum were recorded before executing them; this timing does not constitute preregistration or erase outcome exposure. Inputs are saved participant endpoints, without EEG, weights, new predictions or changes to the original cohort and QC rules.

The fixed comparisons are full minus frozen, full minus LDA-only, and LDA-only minus frozen in the same 12-person primary cohort, plus full minus frozen in the ten participants already eligible for the unflagged sensitivity. For paired differences d_i in BA units, the statistic is $|n^{-1} \sum_i d_i|$. We enumerate all 2^n vectors $s_i \in \{-1, 1\}$ and count $|n^{-1} \sum_i s_i d_i|$ at least as large as the observed value, allowing an absolute boundary tolerance of 10^{-12} BA with zero relative tolerance. The p value is this count divided by 2^n; no Monte Carlo seed or plus-one correction is used for full enumeration.

The enumeration is exhaustive for the sign-flip reference distribution. Its inferential interpretation requires independent participant differences whose signs are exchangeable around zero, conditional on their magnitudes. It is not a randomization test justified by assigned treatment labels, an assumption-free test of a population mean, or evidence that the participant identifiers were independently verified biological identities.

The four tail fractions are 80/4,096, 42/4,096, 1,070/4,096 and 78/1,024. The unadjusted values are 0.01953125, 0.01025390625, 0.26123046875 and 0.076171875; Holm adjustment over exactly this four-test family gives 0.05859375, 0.041015625, 0.26123046875 and 0.15234375. Adjustment

addresses this reported family only and cannot correct all prior analytical exposure. In particular, the primary contrast is not newly declared confirmatory on the basis of its unadjusted value. The sample standardized paired effect for that contrast is $d_z = 0.807128$; percentage-point estimates remain the principal effect measure.

Four assignments in the LDA-only-minus-frozen comparison lie at a floating-point tail boundary. The declared tolerance includes them (1,070/4,096); strict comparison of the saved decimal strings as exact fractions gives 1,066/4,096. Both counts are preserved in the diagnostics, with no change to the rule after calculation. The difference has no material effect on its interpretation.

All original percentile intervals are retained verbatim. Existing CPU code already resampled paired participants jointly; the QC scalar implementation uses the same seed, size and lexicographic order for each relevant contrast. Recomputing with numeric instead of lexicographic participant order can change finite Monte Carlo percentile endpoints slightly, which does not justify replacing the saved results. The unflagged contrast illustrates why interval exclusion and a different paired test are not interchangeable: its retained interval is [0.99, 11.24] points, whereas the sign-flip p value is 0.076171875.

Additional t, signed-rank and sign-test calculations were used only to audit the reviewer's numerical statements, not to choose a favorable test. For tied absolute differences, average-rank sign enumeration and a library option named `exact` need not be equivalent; those audit outputs and the failed initial roundoff-consistency assertion are retained separately. The main article reports only the fixed mean sign-flip checks and their four-test adjustment.

Portable paired inputs, definitions, code and outputs are supplied in S1 Data. The producer requires explicit input and fresh output paths, rejects missing or duplicate participant identities and never invokes a model or an EEG adapter.

485 1. Dataset_Description.pdf. Author-supplied codebook in the TransLearning repository; Available: https://github.com/neurosatya/TransLearning/blob/548470542e63f97ab777684c1f5e4f1cace4b8ed/Dataset_Description.pdf

486 2. Kumar S, Alawieh H, Racz FS, Fakhreddine R, Millán J del R. Transfer learning promotes acquisition of individual BCI skills. PNAS Nexus. 2024;3: pgae076. doi:10.1093/pnasnexus/pgae076

487 3. Correction to: Transfer learning promotes acquisition of individual BCI skills. PNAS Nexus. 2024;3: pgae137. doi:10.1093/pnasnexus/pgae137